COMP2007: Operating Systems & Concurrency
Week 10 – 12:00pm Wednesday – 27 November 2024

valid for 75 minutes from 11:45am
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Figure: Attendance Monitoring

Operating Systems and Concurrency

File Systems 3

COMP2007

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Recap

Last Lecture

1. **User view** of file systems
   - System calls
   - Structures, organisation, file types
2. **Implementation view** of file systems
   - Disk and partition layout
   - File tables
   - Free space management
   - ...

Files

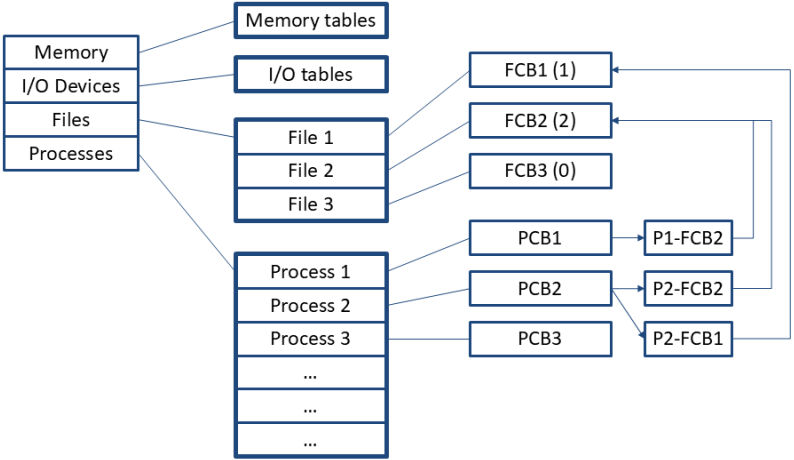
System Calls – Illustration Using “strace” (dtruss on MacOS)

```
# command = strace cat helloWorld.txt > /dev/null

execve("/usr/bin/cat", ["cat", "helloWorld.txt"], /* 34 vars */) = 0
...
open("helloWorld.txt", O_RDONLY)        = 3
...
read(3, "Hello World\n", 1048576)       = 12
write(1, "Hello World\n", 12)           = 12
read(3, "", 1048576)                    = 0
...
close(3)                                = 0
close(1)                                = 0
close(2)                                = 0
exit_group(0)                           = ?
```

File Tables

System Calls



File Storage

How are different blocks belonging to the same file “linked” to the file.

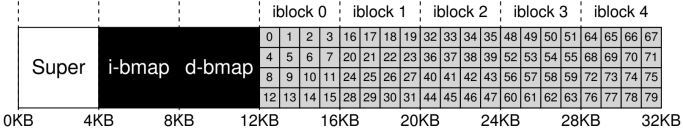


Figure: File Control Blocks

File Storage

How are different blocks belonging to the same file “linked” to the file.

file permissions
file dates (create, access, write)
file owner, group, ACL
file size
file data blocks or pointers to file data blocks

Figure: File Control Blocks

Goals for Today

Overview

- File system **implementations**
  1. Contiguous
  2. Linked lists
  3. File Allocation Table (FAT)
  4. i-nodes (lookups)
- Hard and soft links

File access

Sequential vs. Random Access

- **Files** will be composed of a number of **blocks**
- Files are **sequential** or **random access** (essential for database systems)

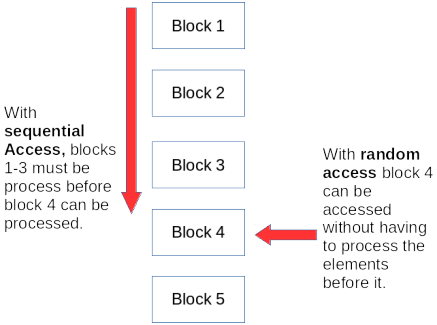


Figure: Types of access to a file

Contiguous Allocation

Concept

- **Contiguous file systems** are similar to **dynamic partitioning** in memory allocation:
  - Each file is stored in a single group of **adjacent blocks** on the hard disk
  - E.g. 1KB blocks, 100KB file, we need 100 contiguous blocks
- Allocation of free space can be done using **first fit**, **best fit**, **next fit**, etc.

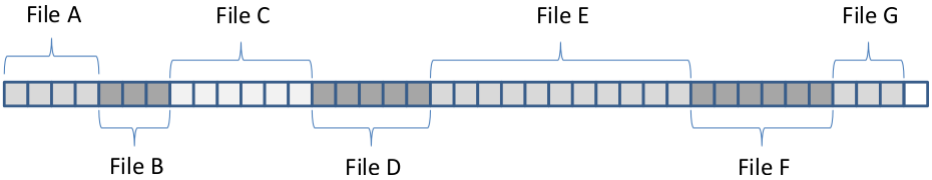


Figure: External Fragmentation when Removing Files

Contiguous Allocation

Advantages

- **Easy to implement**: only location of the first block and the length of the file must be stored (in the FCB)
- **Optimal read/write performance**: blocks are co-located/clustered in nearby/adjacent sectors (seek time is minimised)

File	Start	Length
`req1.c`	0	12
`req1.o`	30	10
`req1`	15	5
`req1.txt`	41	20

Contiguous Allocation

Disadvantages

- **Disadvantages** of contiguous file systems include:
  - The **exact size** of a file (process) is not always known beforehand
  - **Allocation algorithms** are needed to decide which free blocks to allocate (e.g., first fit, best fit)
  - Deleting a file results in **external fragmentation** which requires **de-fragmentation** (which is **slow**)
- Contiguous allocation used for **CD-ROMS/DVDs**
  - External fragmentation is less of an issue since they are **write once**

Contiguous Allocation

Disadvantages

External Fragmentation

How would you resolve external fragmentation?

Linked Lists

Concept

- To avoid external fragmentation, files are stored (discontiguously) in **separate blo** r to paging) that are **linked to one another**
- Only the **address of the first** is stored in the FCB
- Each block contains a **data pointer** to the next block (which takes up space)

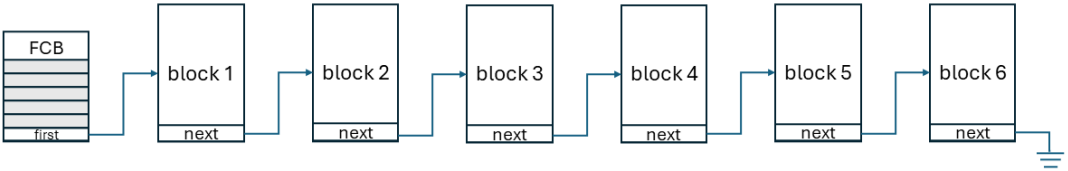


Figure: Linked List File Storage

Linked Lists

Advantages

- **Easy to maintain**: only the first block (address) has to be maintained in the directory entry
- Files can **grow dynamically**: new blocks/sectors are appended at the end
- Similar to paging, there is **no external fragmentation**
- **Sequential access** is straightforward, although **more seek operations** may be required (non-contiguous)

Linked Lists

Disadvantages

- **Internal fragmentation**: the last half of the block is unused (on average)
  - Internal fragmentation will reduce for **smaller block sizes**
- May result in **random (slow) disk access**
  - **Larger blocks** (containing multiple sectors) will improve speed (but increase internal fragmentation)

Linked Lists

Disadvantages (Cont'ed)

Disadvantages

Any other disadvantages?

Linked Lists

Disadvantages (Cont'ed)

- The data within a block is **no longer a power of 2**
- **Reduced reliability**: if one block is corrupt/lost, access to the rest of the file is lost

Linked Lists

Disadvantages (Cont'ed)

Disadvantages

How would you resolve the "power of 2" issue?

File Allocation Tables

Key Concept

- Store the linked-list pointers in a **separate index table**, called a **File Allocation Table (FAT)** (loaded in memory)

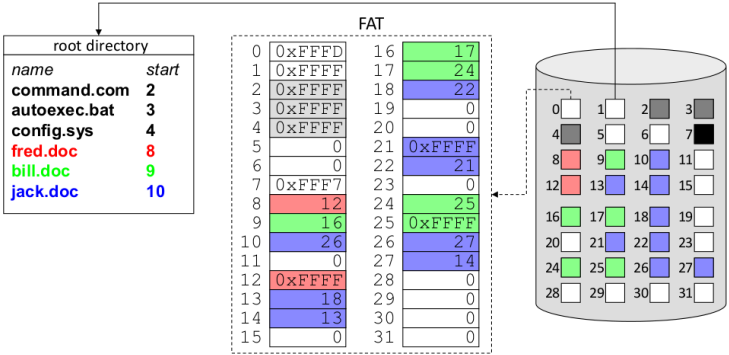


Figure: File Allocation Tables

File Allocation Tables

Advantages and disadvantages

- Advantages:
  - **Block size remains power of 2** (no space is lost in block due to the pointer)
  - **Index table** can be **kept in memory** allowing fast non-sequential/random access (table access itself remains sequential)
- Disadvantages:
  - The **size** of the **file allocation table** grows with the number of blocks/disk size
  - Approximately **200 million entries** for a **200GB disk** with a **1KB block size** (**800MB** at 4 bytes per entry)

File Allocation Tables

Disadvantages

How would you resolve this?

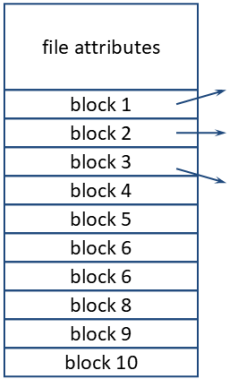
i-nodes

Concept

- Each file has a small data structure (on disk) called **i-node** (index-node) that contains its attributes and block pointers.
  - In contrast to FAT, an i-node is **only loaded when the file is open** (stored in system wide open file table)
  - If every i-node consists of n bytes, and at most k files can be open at any point in time, at most $n \times k$ bytes of main memory are required
- i-nodes are composed of **direct block pointers** (usually 10), **indirect block pointers**, or a combination thereof (e.g., similar to **multi-level page tables**)

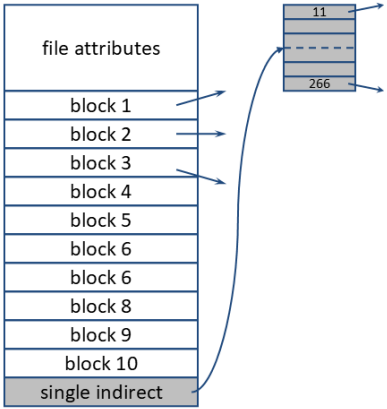
i-nodes

Concept



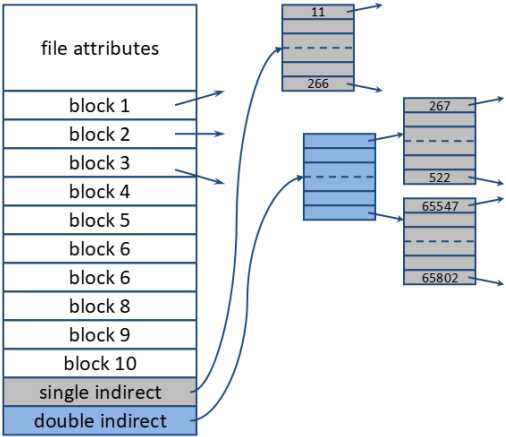
i-nodes

Concept



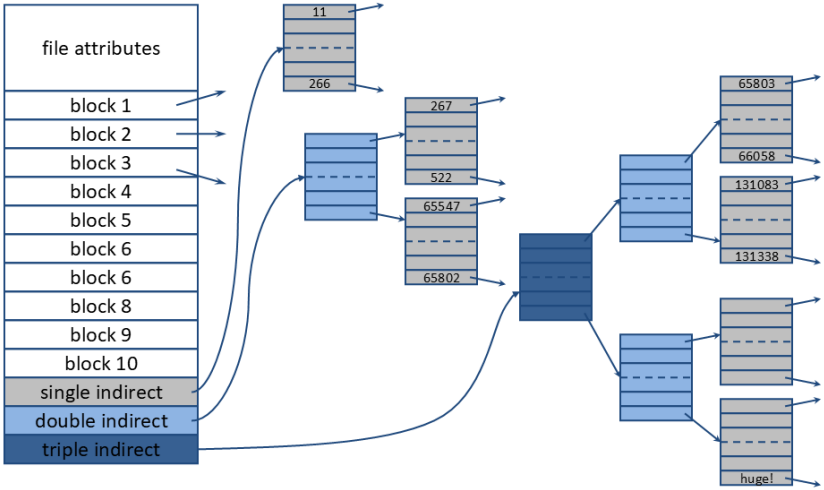
i-nodes

Concept



i-nodes

Concept



i-nodes

Concept

- Assuming a block size of 1KB, and 32-bits disk address space:
  - With **only direct block pointers** the maximum file size is 10KB (1KB $\times$ `NUMBER_OF_DIRECT_BLOCK_POINTERS`)
  - With a **single indirect block** the maximum file size is $(10 + 256) \times 1KB = 266KB$
  - With a **double indirect block** 256 blocks containing 256 pointers each
    - The maximum file size is $(10 + 256 + 256^2) \times 1KB = 65802KB$
  - If we need files larger than this, we will need a triple indirect blocks for a maximum file size of $(10 + 256 + 256^2 + 256^3) \times 1KB$

Directories

Implementation with i-nodes

- In UNIX/Linux/MacOS:
  - All **metadata** for a file (type, size, date, owner, and block pointers) is stored in an **i-node**
  - **Directories** are very simple data structures composed of file name and a pointer to the i-node
- Directories are no more than a special kind of **file**, so they have their own i-node

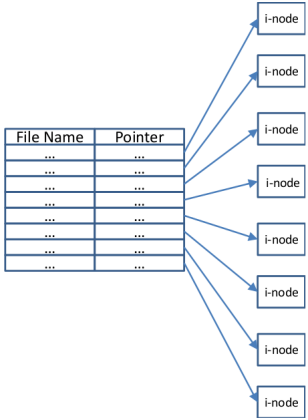


Figure: i-node Directory Structure

File System Comparison

Contiguous vs. Linked vs. Indexed

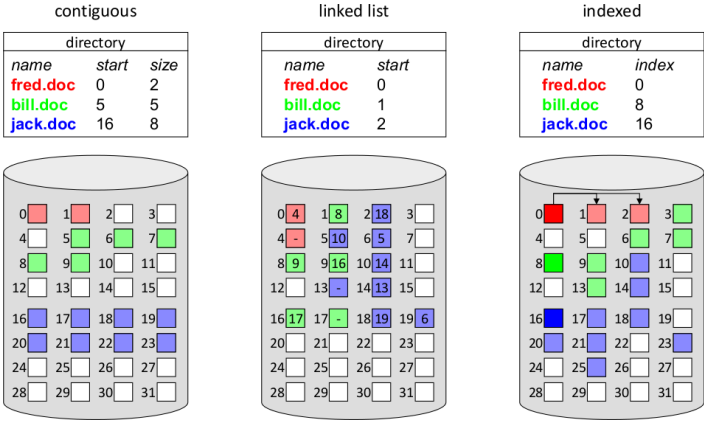


Figure: Contiguous vs. Linked List (or FAT) vs. i-nodes (or indexed)

i-nodes

Lookups

- **Opening a file** requires the disk blocks to be located
  - **Absolute file names** are located relative to the root directory
  - **Relative file names** are located based on the current working directory
- E.g. Try to locate `/usr/gdm/mbox`

/ inode

1
size
mode
times
2

/ contents

2	
1	.
1	..
4	bin
7	dev
14	lib
9	etc
6	usr
8	tmp

/usr inode

6
size
mode
times
132

/usr contents

132	
6	.
1	..
19	fred
30	bill
51	jack
26	gdm
45	cfi

/usr/gdm inode

26
size
mode
times
406

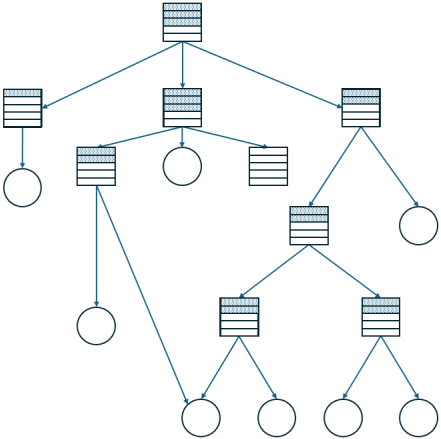
/usr/gdm contents

406	
26	.
6	..
64	research
92	teaching
60	mbox
17	grants

Figure: Locating a File

i-nodes: Sharing files between directories

Hard and Soft Links



i-nodes: Sharing files between directories

Hard and Soft Links

- There are two approaches to **share a file**, e.g. between directory B and C, where C is the 'real' owner:
  - **Hard links**: maintain two (or multiple) references to the same i-node in B and C
    - the i-node link reference counter will be set to 2
  - **Symbolic links**:
    - The owner maintains a reference to the i-node in, e.g., directory C
    - The "referencer" maintains a small file (that has its own i-node) that contains the location and name of the shared file in directory C
- What is the **best approach**? $\Rightarrow$ both have advantages and disadvantages

i-nodes: Sharing files between directories

Hard Links

- Hard links are the **fastest way of linking files**!
- Disadvantages of hard links:
  - Assume that the owner of the file **deletes** it:
    - If the i-node is also deleted, any hard link will, in the best case, **point to an invalid i-node**
    - If the i-node gets **deleted** and "**recycled**" to point to an other file, the hard links will **point to the wrong file**!
  - The only solution is to **delete the file**, and **leave the i-node intact** if the "**reference count**" is larger than 0 (the original owner of the file still gets "charged" for the space)

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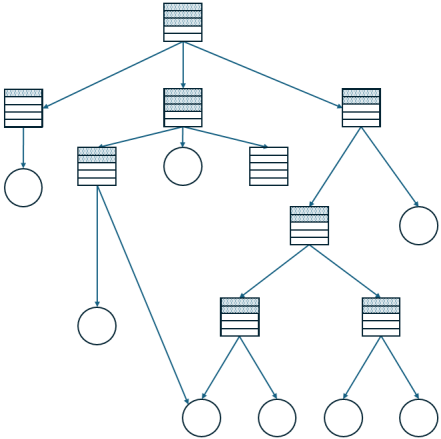
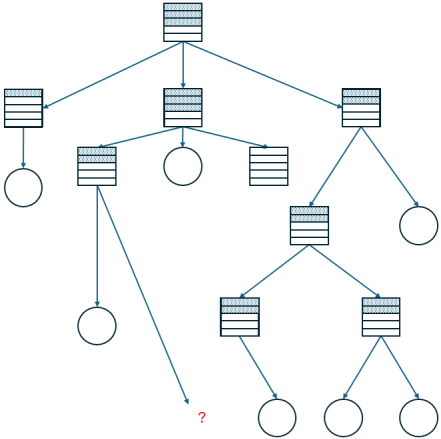
i-nodes: Sharing files between directories

Deleting Files



i-nodes: Sharing files between directories

Deleting Files



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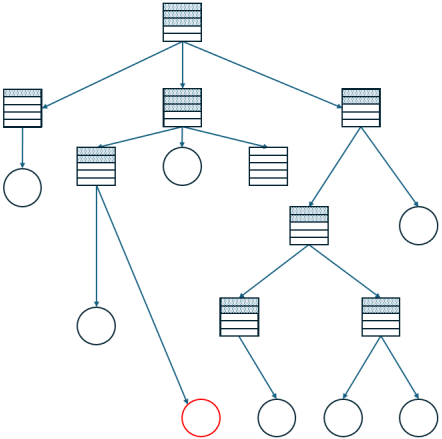
i-nodes: Sharing files between directories

Deleting Files



i-nodes: Sharing files between directories

Hard Links

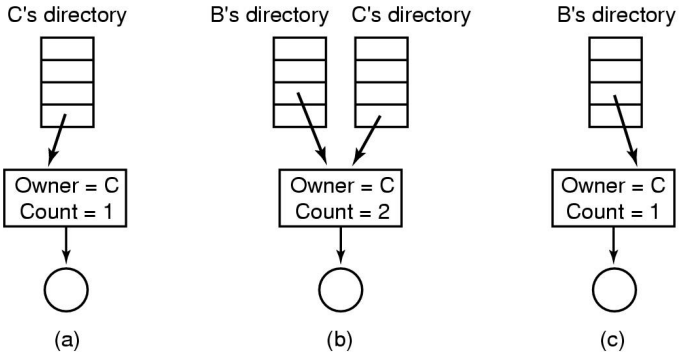


Figure: (a) Before creating a link. (b) After creating a link. (c) after the orignal owner removes the file (Tanenbaum)

i-nodes

Soft Links

- Disadvantages of soft links:
  - They result in an **extra file lookup** (once the link file has been found, the original file needs to be found as well)
  - They require an **extra i-node** for the link file
- Advantages of symbolic links:
  - There are **no problems with deleting** the original file $\Rightarrow$ the file simply does not exist any more
  - They can **cross the boundaries of machines**, i.e. the linked file can be located on a different machine

i-nodes

Hard and Soft Links in Unix

```
[pszgd@severn ~]$ pwd
/home/pszgd
[pszgd@severn ~]$ ls -i labs/
3250013 req1b.c
[pszgd@severn ~]$ ln labs/req1b.c hardLink
[pszgd@severn ~]$ ln -s labs/req1b.c softLink
[pszgd@severn ~]$ ls -ali hardLink softLink
3250013 hardLink
3250021 softLink -> labs/req1b.c
[pszgd@severn ~]$ rm labs/req1b.c
```

File System Examples

Unix vs. Windows

- The Unix V7 File System:
  - **Tree structured** file system with links
  - Directories contain **file names** and **i-node numbers**
  - i-nodes contain **user and system attributes** (e.g. count variable)
  - One single, double, and triple **indirect blocks** can be used
- More sophisticated File Systems were developed later (e.g. ext3/4)
- Windows:
  - Up to XP used FAT-16 and FAT-32
  - From XP moved to NTFS (64 bits) because of file size limitations
    - NTFS uses File Tables, with bigger i-nodes that can also contain small files and directories
  - More recently uses ReFS

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Recap

Take-Home Message

- Contiguous, linked list, FAT and i-nodes as file system implementations.
- Lookups with i-nodes.
- Hard and Soft Links

Test Your Understanding

File Systems

Exercises

With i-nodes, the maximum file size that we can have depends on the block size and the number of indirections.

- Assuming a 32-bit disk address space, what is the maximum (theoretical) file size for a FAT file system with a drive of 500GB and a block size of 1KB? (without accounting for directory metadata)
- The most used implementation of FAT is known as FAT-32. Investigate why there is a theoretical limitation of 4GB per file (and sometimes even less than 2GB).

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